Effects of a 20-Min Nap Post Normal and Jet Lag Conditions on P300 Components in Athletes

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ABSTRACT

Post-lunch sleepiness belongs to biological rhythms. Athletes take a nap to counteract afternoon circadian nadir, in prevision of disturbed sleep. This study examined the effects of brief post-lunch nap on vigilance in young and healthy athletes. The P300 components, physiological and cognitive performances were assessed either after nap or rest, following a night of normal sleep (NSC) or simulated jet lag condition (5-h advance-JLC). P300 wave is the positive deflection at about 300 ms in response to a rare stimulus, representing higher information processing. P300 amplitude reflects the amount of attention allocated whereas P300 latency reflects time spent on stimulus classification. P300 amplitude was significantly increased (Fz:11.14 ± 3.0vs9.05 ± 3.2 µV; $p < 0.05$) and P300 latency was shorter (Pz:327.16 ± 18.0vs344.90 ± 17.0 ms; $p < 0.01$) after nap in NSC. These changes were accompanied by lower subjective sleepiness (19.7 ± 9.6vs27.5 ± 16.5; $p < 0.05$) and decrease in mean reaction times (MRT: divided attention, 645.1 ± 74.2vs698 ± 80.4 ms; $p < 0.05$). In contrast, in JLC, only P300 amplitudes (Fz:10.30 ± 3.1vs7.54 ± 3.3 µV; $p < 0.01$ and Cz: 11.48 ± 3.1vs9.77 ± 3.6 µV; $p < 0.05$) increased but P300 latencies or MRT did not improve. These results indicated improvements in speed of stimulus evaluation time. Napping positively impacts on cognitive processing, especially when subjects are on normal sleep schedules. A nap should be planned for athletes whose performance requires speedy and accurate decisions.

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Introduction

A number of circumstances in the lives of athletes may result in psychophysiological and sleep disturbances. Among these circumstances, several are frequently encountered, such as changes in environment, early starts of athletic competitions, time zone changes after intercontinental flights, all factors that may produce anxiety, affect mood [10, 14], cause some cognitive decline, (for review, see [29]), increase fatigue, drowsiness and errors in performance involving cognitive processes [34]. These factors may be potent agents of stress and may lead athletes to take a brief nap to compensate the sleep debt, sustain alertness and arousal state, and improve physical and cognitive performance [3, 46]. In these situations, it remains unclear whether it is better to propose a brief nap or a rest without sleeping for athletes keeping their usual sleep schedules or undergoing a jet lag so that vigilance and alertness are maintained and optimal cognitive performance achieved.

Previous studies have demonstrated that a 20-min nap is most often recommended to restore alertness because it results in min-

imal sleep inertia and contains mostly stage 2 non-REM sleep. Nonetheless, in a previous study, Petit et al. [35, 36] reported that this type of nap between 13:00 and 14:00 h or 08:00 and 09:00 h in non-shifted and shifted conditions respectively did not improve anaerobic physical performance but modified the circadian rhythm parameters. Conversely, research has consistently shown that daytime napping improves reaction times, attention [5, 7, 39, 42], memory consolidation [33] and enhances physiological alertness [6, 20, 22, 43]. An improvement in alertness, associated with a decrease in sleepiness has also been reported by Waterhouse et al. [46] in response to a brief nap (30 min), which took place after partial sleep deprivation. Nevertheless, in most of these studies, alertness was assessed only by visual analogic scales, either in healthy and young subjects [25, 40, 41] or in elderly people [26]. There is little information about cognitive function and vigilance in athletes using neurophysiological techniques such as event-related potentials (ERP).

ERP are brain potentials recorded following the presentation of stimuli that represent various stages of information processing [37]. Event-related potentials reflect neuro-electric activity related to components of cognition such as attention, perception, reaction time, short-term memory, but also enable estimation of the level of vigilance and acute sleep inertia [2, 41]. The wave P300 is the positive deflection at about 300 ms in response to a rarely occurring stimulus, representing higher cognitive functions of information processing, working memory, or stimulus categorization [45]. P300 amplitude reflects the amount of attention allocated whereas the latency of the P300 reflects the time spent on stimulus classification.

Several studies have provided evidence that ERP are sensitive to changes in quantity [18, 24, 49] and quality of sleep [12], and to changes in attention and arousal levels (for review see [8, 11]). Conversely, the impact of sleep obtained through a short nap on vigilance and alertness measurements in athletes is poorly documented.

We hypothesize that the cognitive improvements induced by a short nap taking place after normal sleep schedules or simulated jet lag are consequences of neurophysiological changes in brain processes that would lead to higher P300 amplitude, shorter P300 latency and greater alertness, so that better performance is achieved by athletes.

Thus, the first aim of this study was to examine the effects of 20-min nap on cognitive performance as measured by event-related potentials (ERP), subjective alertness and psychomotor performance in highly physically trained subjects keeping their usual sleep schedules. The second aim was to investigate whether a 20-min nap may have different effects after simulated jet lag.

Methods

Participants

Twenty young highly trained male athletes were recruited, of whom sixteen participated in the study. Anthropometric and circadian characteristics are detailed in ▶ **Table 1**. Four subjects were excluded because of reported sleep disturbances. Two subjects with poor quality evoked potential data were excluded for Even Related Potentials data processing. Enrolled subjects did not report any significant previous or existing health problems, including substance abuse, and specifically, they did not have any sleep related complaints or partial sleep restriction. The inclusion criteria about sleep wake habits and circadian typology have been precisely described in previous publications [36–35]. None of them habitually napped.

▶ **Table 1** Anthropometric and circadian characteristics of subjects.

Anthropometric characteristics	
Age (years)	22.20 ± 1.7
Height (cm)	178.30 ± 5.6
Body mass (kg)	73.60 ± 7.9
Fat mass (%)	12.60 ± 3.4
$\dot{V}O_2$ max ($mL.min^{-1}.kg^{-1}$)	55.50 ± 9.1
Circadian characteristics	
Chronotype (ua)	52.88 ± 5.4
TST/night (min)	491.44 ± 41.1
SOL (min)	15.75 ± 12.1
SE (%)	94.81 ± 3.0

Values are expressed in means ± SD. Chronotype is measured from Horne and Ostberg « morningness-eveningness » questionnaire. TST: total sleep time, SOL: sleep onset latency, SE: sleep efficiency = total sleep time estimated / time in bed × 100. n = 16.

Ethics

The study meets the ethical standards of the journal [19]. Local ethics committee approved the study (Protocol number: T/2009/06 "WINAP"). All volunteers provided written informed consent and were informed about their right to withdraw from the study at any time.

Experimental design

The experimental design (sleep conditions, polysomnography recordings and scoring, post-lunch nap and rest recordings, core body temperature measurements) are described in detail in ▶ **Fig. 1** and has previously been described in detail in Petit et al. [35].

All nights took place in a sleep laboratory, in constant temperature (18–20 °C), noise (< 40 dB), and dim light (< 10 lux) conditions. The phase advance condition comprised a 5-h phase advance night during which lights were turned off from 18:00 h to 02:00 h the day after. Participants were instructed to arrive at the laboratory at 14:00 h for dinner before the night, and sleep was shifted in advance of phase by 5 h. Subjects were asked to change their clocks and watches. They went to bed at 18:00 h and were woken at 02:00 h. During these nights, subjects were accommodated in a quiet room and slept in a comfortable, soundproofed bedroom. Once subjects woke up, they were exposed to 5 h of room light (Pharos Max, 10000 Lux – 76 cm) to stay awake.

All the subjects spent five nights in a sleep laboratory. All participants first spent one habituation night in the laboratory. Then subjects were randomized in a cross-over designed study for the remaining four nights: one normal night followed by a rest without nap, one normal night followed by a nap, one phase advance night followed by a rest without nap, and one phase advance night followed by a nap.

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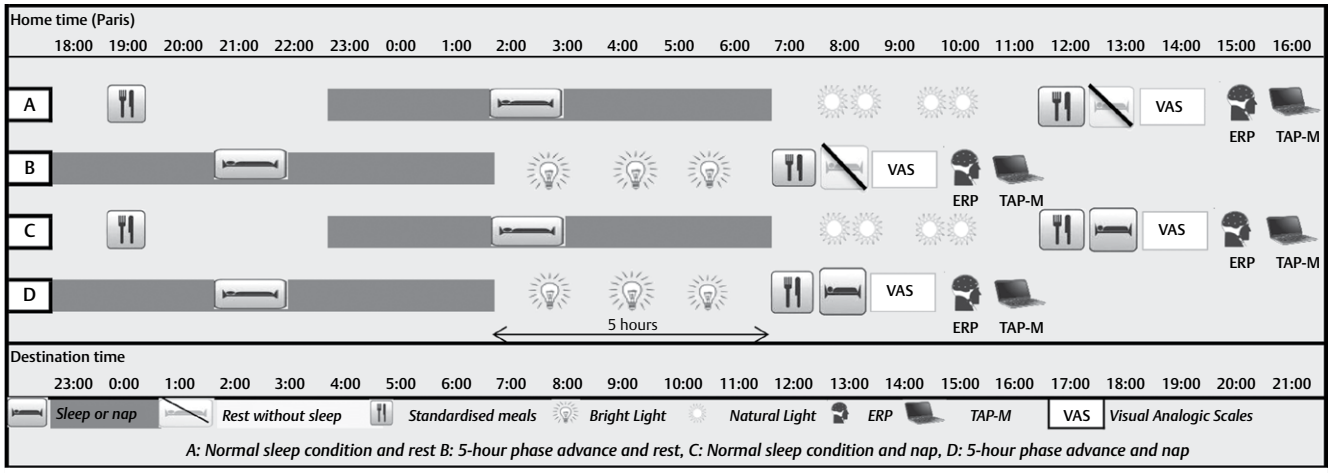

▶ **Fig. 1** Experimental protocol.

Sleep recordings

A camera constantly monitored subjects and the EEG recorded was transmitted in real-time to a computer manned by one of the experimenters. Sleep was assessed using standard polysomnography techniques (PSG) using an 8-channel EEG device (i. e. F4, C4, 02, M2, F3, C3, 01, M1 electroencephalogram, right and left electrooculogram, electromyogram). PSG data were recorded directly to a data acquisition, storage, and analysis system (Schwarzer GmbH, Munich, Germany; OSG BrainRT, OSG Bvba, Rumst, Belgium). The EEG recordings from nights were scored using standardized criteria (Rechtschaffen & Kales, 1968) to obtain the overnight pattern of sleep stages according to the AASM standard rules (Iber C et al., 2007). EEG monitoring from the mornings were screened to check for micro sleeping.

We recorded 9 sleep parameters: total sleep time (TST, calculated between time of sleep onset – computed after 3 epochs of N1 or 1 epoch of N2 – and time of awakening, min), sleep efficiency (SE = TST/Time in Bed * 100, %), number of awakenings (n), sleep onset latency (SOL, min), total time in stage 1 (TTN1, min), total time in stage 2 (TTN2, min), total time in stage 3 (TTN3, min), total time in stage REM (rapid eye movement sleep; TTREM, min), and quality of sleep (calculated by the ratio (TT N3 + TT REM)/TST %).

Nap recordings

Naps were scheduled at 13:00 after normal sleep condition **(C)** and at 08:00 in phase-advance condition **(D)**. They were continuously monitored and scored by PSG. Subjects were required to lie on a bed in a darkened room for one hour and were awakened by intercom when about 20 min of sleep had elapsed from 3 epochs of stage 1 or 1 epoch of stage 2.

In rest conditions without nap (**A** and **B**), subjects remained lying in a bed in the same conditions as in nap conditions, for one hour, between 13:00 and 14:00 and 08:00 and 09:00 in normal sleep and phase-advance conditions respectively. They were asked not to take a nap. In case of any signs of sleep, they were awakened by intercom. EEG recordings confirmed that they did not sleep throughout this period. Subjects were not submitted to darkness, contrary to nap condition.

Alertness assessments

After measuring subjective alertness by Visual Analogic Scales (VAS), participants performed an auditory event-related potentials (ERP) test, performed a battery of tests for Attentional Performance (TAP-M) 30 min after their nap or rest without sleeping.

Auditory Event-related potentials (ERP)

The P300, an event-related potential that is a neurophysiological correlate of cognitive function, was measured. The auditory oddball task involved differentiating between tones of two different pitches. This task involved delivering a 70 dB tone binaurally (duration = 50 ms; interstimulus interval = 1000–2000 ms). The total number of trials delivered per session was 400. The standard was an 800 Hz tone presented on 80 % of trials and the target was a 1600 Hz tone presented at random on 20 % of trials. The participants' task was to respond as quickly and accurately as possible to the rare, target tones by pressing a hand-held button. P300 amplitudes and latencies in particular were studied. Reaction time to each correct trial was stored on computer, but not displayed to participants as feedback.

Electro-encephalogram (EEG) recordings and data analysis

EEG was recorded directly using an Electro cap (Electro-Cap International Inc.), and a data acquisition, storage, and analysis system (Schwarzer GmbH, Munich, Germany; OSG BrainRT, OSG Bvba, Rumst, Belgium). A 21-channel EEG system was used, with a 19-channel tin electrode cap with electrodes Fp1, Fp2, Fz, F3, F4, F7, F8, Cz, C3, C4, T7, T8, Pz, P3, P4, P7, P8, A1, A2, fitted according to the International 10-20 system, and the cap electrodes were referenced to earlobe electrodes. The ground electrode was placed on the forehead (between the Fp1/2 and Fz electrodes). EOG electrodes were placed 1 cm from the outer canthus of each eye, while EMG was recorded via a bipolar channel placed submentally. Impedance was kept below 5 kΩ. A 1 Hz low pass cut-off and a 35 Hz high pass cut-off software filters were applied to record EEG signals during the oddball task, and prior to ERP analyses, and the sampling rate was 500 Hz. A 50 Hz notch filter was applied for recording. Moreover, since EEG channels can be contaminated by multi-

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ple sources such as eye blinks, movement, and muscle activity, a threshold was also applied to all EEG channels online during data acquisition. It has been specified that the trial spectra should not deviate from the mean by ± 50 dB in the 0-2 Hz frequency window (for catching eye movements) and should not deviate by + 25 or -100 dB in the 20–40 Hz frequency window (for detecting muscle activity). Trials with omission and commission errors were excluded from averaging (29.6 ± 3.5 %).

As detailed by Lopez-Calderon and Luck [30], ERP data were analyzed by EEGLAB toolbox, an open-source toolbox for processing and analyzing ERP data in the MATLAB environment (The Mathworks, Inc., Natick, MA, USA). EEG data recorded during the auditory ERP task were divided offline into discrete trials. Each window began 200 ms prior to stimulus onset and continued until 800 ms after stimulus presentation. If the automatic averaging routine resulted in too many trials being excluded (more than 30 % of trials), then individual trials were visually inspected and included. Then there was no obvious eye movement artifact on the EEG in the interval overlapping with ERPs of interest.

Analyses for the P300 components were run for the non-target and target stimuli. Trials were averaged separately for non-target and target stimuli for each individual. The amplitudes and latencies of the raw ERP components were measured in the three-midline electrodes (Fz, Cz, and Pz) where they are maximal. P300 was measured between 250 and 390 ms. For the P300 components, amplitude was measured at the maximum local peak in the defined time interval [31] and latencies as the time from stimulus onset to the point of maximum positive amplitude within the time window of interest. For target stimuli, mean reaction times (RT), as well as the average of non-reactions, correct and incorrect responses were calculated.

Subjective alertness measurement

Subjective alertness was measured using a 100 mm visual analogue scale (VAS) [4]. This scale required participants to rate how alert they felt, by marking on a non-numeric 100 mm line anchored with "Awakened" on one end (i. e. 0 mm) and "Sleepy" on the other end (i. e. 100 mm). These variables are used in order to assess both lapsing and response slowing which are known to accompany sleepiness [44].

Test of Attentional Performance (TAP-M)

Subjects performed the battery of tests for Attentional Performance (TAP, version 2.2, Psytest, Herzogenrath, Germany) [48]. The TAP battery consisted of six tests based on reaction time measurements, namely: alertness, divided attention, sustained attention, visual scanning, flexibility and distractibility. Following a familiarization session, subjects performed one or two trials, according to their needs to ensure comprehension of the task demands and to avoid a learning effect. Participants sat comfortably in front of a computer screen and responded to each stimulus by pressing a button connected to the computer with the forefinger of their dominant hand. Subjects were instructed to perform the tasks as quickly as possible while maintaining a high level of accuracy. Low values in mean reaction times indicate good performance.

Statistical analysis

Statistical analyses were performed using Stata/SE 10.1 for Windows (Stata Corp, College Station, TX, USA). Quantitative data are expressed as mean ± standard deviation (SD). A p value < 0.05 was considered statistically significant. The effects of nap and phase-advance were analyzed by two-way ANOVA measures, for polysomnographic variables during the nights and naps, P300 parameters, psychomotor performance variables, and core body temperature values. In case of nap effect, a Newman–Keuls post hoc test was performed to compare the data between conditions with nap vs without nap. The paired Student t test for continuous variables with normal distribution or Wilcoxon paired test for continuous variables without normal distribution were then used, respectively, to evaluate differences in ERP data (amplitudes, latencies), subjective alertness in VAS, mean reaction times in 6 fields of attention of TAP-M and core body temperature parameters.

Results

Polysomnographic variables during the nights

Sleep architecture was identical in normal sleep condition (A vs C), in 5-h phase advance condition, followed, or not by a nap (B vs D). The phase advance affected the composition of sleep compared with that observed during a normal night (A vs B). TST was significantly lower compared to the normal sleep condition (365.10 ± 106.30 vs 449.60 ± 18.50 min; $p < 0.01$). SE ($p < 0.01$), SOL ($p < 0.01$), TTN2 ($p < 0.01$), TTREM ($p < 0.001$) and % REM ($p < 0.01$) were also significantly reduced by the 5-h phase advance. TTN1 and TTN3 were unchanged and quality of sleep was not altered (▶ **Table 2**).

Polysomnographic variables during naps

The composition of post-lunch nap was similar after normal night or phase advance night. The subjects slept for 21–22 min in the hour they stayed in bed with sleep efficiency that approximated 70 %. Percentages of stage 2 (N2) and 3 (N3) were predominant (▶ **Table 3**). Only the number of awakenings decreased in phase advance condition ($p < 0.05$). Rapid eye movement (REM) sleep occurred after normal night and phase advance night without being statistically significant between both conditions ($p < 0.08$).

Alertness assessments

Auditory Event related potentials (ERP)

Whatever the conditions, napping improved the vigilance performance. In normal sleep condition P300 amplitude increased in Fz ($p < 0.05$) and P300 latency decreased in Pz ($p < 0.01$) compared to the values obtained without nap (▶ **Table 4**; ▶ **Fig. 2**). In the 5-h phase advance condition, only P300 amplitudes increased in Fz ($p < 0.01$) and Cz ($p < 0.05$) compared to the values recorded without a nap.

Subjective alertness (VAS)

Higher values on visual analogic scales indicated a more sleepy state, whereas low values indicated a more awakened state. In the normal condition, the awakened sensation was greater with a nap than without a nap (19.7 ± 9.6 vs 27.5 ± 16.5; $p < 0.05$).

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▶ **Table 2** Night polysomnographic variables in normal sleep and 5-h phase advance conditions without (A/B) or with nap (C/D).

n = 16	Phase advance effect	Nap effect	Phase advance x Nap effect	Normal sleep conditions			5-h phase-advance conditions			A vs B	C vs D
	p	p	p	Without nap (A)	With nap C)	p	Without nap (B)	With nap (D)	p	p	p
TST (min)	<0.01	ns	ns	449.60 ± 18.50	446.60 ± 16.60	ns	365.10 ± 106.30	367.80 ± 87.20	ns	<0.01	<0.01
SE (%)	<0.001	ns	ns	92.00 ± 3.60	92.30 ± 3.40	ns	73.10 ± 20.80	74.30 ± 18.80	ns	<0.01	<0.001
Wake (n)	ns	ns	ns	22.50 ± 8.70	21.50 ± 6.69	ns	20.06 ± 7.96	20.06 ± 9.07	ns	ns	ns
SQ (%)	ns	ns	ns	60.69 ± 9.45	62.76 ± 8.79	ns	58.32 ± 10.57	62.38 ± 9.19	ns	ns	ns
SOL (min)	<0.01	<0.01	<0.05	13.45 ± 8.01	16.91 ± 9.77	ns	7.41 ± 3.65	7.56 ± 6.37	ns	<0.01	<0.001
TTN1 (min)	<0.05	ns	ns	8.34 ± 8.11	5.03 ± 5.51	ns	12.66 ± 16.11	6.78 ± 8.05	ns	ns	ns
TTN2 (min)	<0.01	ns	ns	168.40 ± 43.30	160.60 ± 36.40	ns	134.50 ± 55.60	130.60 ± 47.60	ns	<0.01	<0.05
TTN3 (min)	ns	ns	ns	170.31 ± 41.35	184.72 ± 37.83	ns	157.50 ± 56.60	166.72 ± 45.82	ns	ns	ns
TTREM(min)	<0.001	ns	ns	96.80 ± 24.40	96.50 ± 20.90	ns	54.50 ± 29.80	59.90 ± 27.60	ns	<0.001	<0.001

Values are expressed in means ± SD. TST: total sleep time; SE: sleep efficiency (%); Wake: number of awakenings; SQ: sleep quality = total time in stage 3 + REM/TST; SOL: sleep onset latency; TT: total time in each stage. REM: rapid eye movement. ns: non-significant.

▶ **Table 3** Nap polysomnographic variables after normal sleep and 5-h phase advance nights.

n = 16	After normal sleep night	After 5-h phase advance night	p
TST (min)	21.72 ± 2.56	21.29 ± 1.58	ns
SE (%)	70.73 ± 14.29	70.44 ± 21.06	ns
Wake (n)	2.00 ± 2.71	1.13 ± 1.31	p < 0.05
SOL (min)	7.69 ± 4.14	7.94 ± 9.17	ns
TTN1 (min)	2.19 ± 2.06	1.53 ± 1.42	ns
N1 (%)	9.89 ± 7.15	9.49 ± 6.89	ns
TTN2 (min)	11.56 ± 5.49	12.14 ± 4.65	ns
N2 (%)	52.20 ± 20.82	56.01 ± 22.33	ns
TTN3 (min)	7.78 ± 5.22	7.09 ± 5.65	ns
N3 (%)	36.80 ± 25.47	32.06 ± 25.41	ns
TTREM (min)	0.19 ± 0.75	0.53 ± 1.36	p < 0.08
REM (%)	1.11 ± 4.40	2.44 ± 5.58	p < 0.08

Values are expressed in means ± SD. TST: total sleep time; SE: sleep efficiency; Wake: number of awakenings; SQ: sleep quality = (total time in stage 3 + REM)/TST; SOL: sleep onset latency; TT: total time in each stage. REM: rapid eye movement. % is expressed in total sleep time. ns: non-significant.

In the 5-h phase advance condition, there was no significant difference in subjective sensations (awakened-sleepy) after a nap compared to data measured without a nap.

Test of Attentional Performance (TAP-M)

In TAP-M, low values in mean reaction times (simple and complex tasks), indicate good performance. In the normal sleep condition, a nap led to minor modifications in psychomotor performance. Indeed, only better reaction times in divided attention were recorded ($p < 0.05$), compared to values obtained without a nap (▶ **Table 5**). When the subjects were shifted by 5-h, napping improved neither simple nor complex tasks, compared to values recorded without a nap.

Core body temperature

All core body temperatures data are presented in ▶ **Table 6**. Times are expressed in local time. In normal sleep conditions, amplitude, zenith and mesor significantly increased by 0.07 °C ($p < 0.05$), 0.20 °C ($p < 0.001$) and 0.11 °C ($p < 0.001$) respectively, in nap vs no-nap condition. In 5-h phase advance condition, period, amplitude, zenith, nadir and mesor were similar in nap and no-nap conditions. Nevertheless, acrophase and batyphase were delayed 59 and 47 min, respectively, in the nap condition compared to the no-nap condition ($p < 0.01$).

Discussion

To the best in our knowledge, this study is the first to evaluate in athletes the effects of a 20-min nap scheduled during the post-prandial period on subsequent neurophysiological processes (P300 components measured in Cz, Pz and Fz electrodes), subjective alertness and psychomotor performance in normal sleep matching home sleep or 5-h phase advance conditions simulating eastward transmeridian travel. An ERP task was used to provide neuro-

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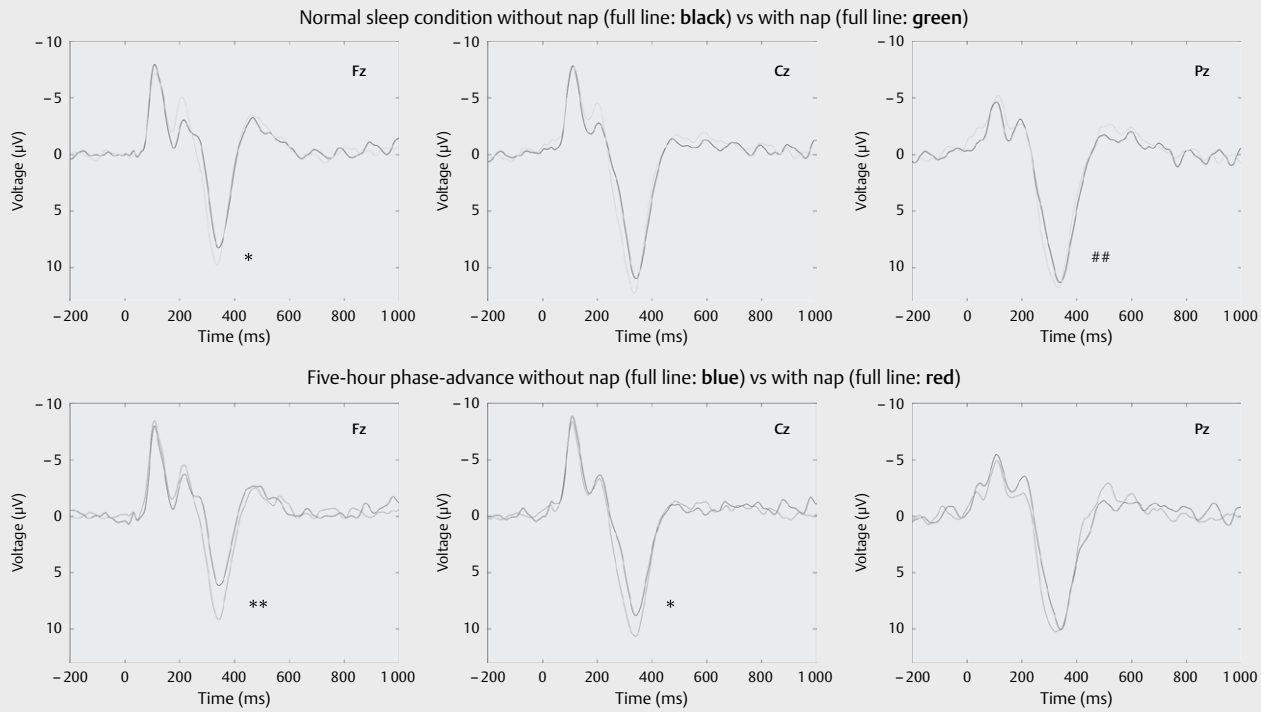
▶ **Table 4** P300 amplitudes and latencies on Fz, Cz, Pz derivations after normal sleep and 5-h phase advance nights without (A/B) or with nap (C/D).

n = 14	Phase advance effect	Nap effect	Phase advance x Nap effect	Normal sleep conditions			5-h phase advance conditions		
	p	p	p	Without nap (A)	With nap (C)	p	Without nap (B)	With nap (D)	p
Amplitudes (μV)									
Fz	ns	< 0.01	< 0.01	9.05 ± 3.2	11.14 ± 3.0	< 0.05	7.54 ± 3.3	10.30 ± 3.1	< 0.01
Cz	ns	ns	< 0.05	11.63 ± 4.0	12.55 ± 3.7	ns	9.77 ± 3.6	11.48 ± 3.1	< 0.05
Pz	ns	ns	ns	12.21 ± 3.6	12.38 ± 3.4	ns	11.00 ± 3.6	11.37 ± 2.9	ns
Latencies (ms)									
Fz	ns	ns	ns	345.48 ± 18.1	335.88 ± 18.6	ns	346.30 ± 21.4	337.77 ± 17.2	ns
Cz	ns	ns	ns	344.61 ± 16.7	334.50 ± 12.4	ns	340.55 ± 27.0	337.92 ± 15.0	ns
Pz	ns	< 0.01	ns	344.90 ± 17.0	327.16 ± 18.0	< 0.01	331.78 ± 21.8	322.72 ± 31.7	ns

Values are expressed in means ± SD. Fz: frontal electrode site, Cz: central electrode site, Pz: parietal electrode site. ns: non-significant.



▶ **Fig. 2** Grand average for P300 amplitudes and latencies for Fz, Cz, Pz derivations after normal sleep and 5-h phase advance nights without (A/B) and with nap (C/D).

physiological data on information processing. A TAP M test was performed to examine psychomotor performance, which represents a robust measure sensitive to changes in alertness [15].

The main results of the study indicated that a 20-min nap between 13:00 and 14:00 h after a night of normal nocturnal sleep increased P300 amplitude in frontal derivation and decreased P300 latency in parietal derivation elicited in an auditory oddball paradigm. These changes are accompanied by a lower subjective sleepiness and better mean reaction times in divided attention. In contrast, a nap which took place between 08:00 and 09:00 h after a night simulating 5-h jet lag, only increased P300 amplitudes in frontal and central derivations without modifying P300 latencies or mean reaction times in attentional tests.

Discovery of the P300 event-related potential (ERP) stimulated the use of brain recording methods to assess human cognition. The P300 wave is commonly elicited in an oddball paradigm when a subject detects an occasional target stimulus in a regular train of standard stimuli. Its amplitude varies with the improbability of the target. Its latency varies with the difficulty of discriminating the target from standard stimuli. It may represent an index information-processing speed. P300 occurs when the attention-driven stimulus signal is transmitted to temporal and parietal structures.

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▶ **Table 5** Psychomotor performance in 6 fields of attention after normal sleep and 5-h phase advance nights without (A/B) or with nap (C/D).

n = 16	Phase advance effect	Nap effect	Phase advance x Nap effect	Normal sleep conditions			5-h phase advance conditions		
	p	p	p	Without nap (A)	With nap (C)	p	Without nap (B)	With nap (D)	p
Phasic alertness	ns	ns	ns	218.37 ± 15.9	212.62 ± 16.3	ns	215.25 ± 17.4	223.43 ± 21.6	ns
Flexibility	ns	ns	ns	550.68 ± 101.1	542.18 ± 132.8	ns	529.62 ± 144.6	553.12 ± 141.4	ns
Divided attention	ns	< 0.05	ns	698.00 ± 80.4	645.10 ± 74.2	< 0.05	696.37 ± 92.3	693.90 ± 74.1	ns
Visual scanning	ns	ns	ns	3604.87 ± 1061.1	3262.50 ± 587.0	ns	3213.27 ± 846.2	3600.33 ± 725.7	ns
Distractibility	ns	ns	ns	406.62 ± 49.0	391.00 ± 43.3	ns	418.25 ± 48.4	405.93 ± 36.1	ns
Sustained attention	ns	ns	ns	474.87 ± 65.9	439.90 ± 65.2	ns	483.68 ± 75.4	473.60 ± 53.8	ns

Values are expressed in means ± SD. Reaction times in fields of attention are all expressed in ms. ns: non-significant.

▶ **Table 6** Cosinor analysis of core body temperature data in normal sleep and 5-h phase advance conditions without (A/B) or with nap (C/D).

	Phase advance effect	Nap effect	Phase advance x Nap effect	Normal sleep conditions			5-h phase advance conditions		
n = 16	p	p	p	Without nap (A)	With nap (C)	p	Without nap (B)	With nap (D)	p
Period (hours)	ns	ns	ns	24.71 ± 0.81	24.81 ± 0.73	ns	24.61 ± 0.74	24.34 ± 0.98	ns
Acrophase (hours)	< 0.001	< 0.01	ns	16:42 ± 00:52	17:07 ± 00:20	ns	13:33 ± 00:25	14:32 ± 1:06	< 0.01
Batyphase (hours)	< 0.001	0.07	< 0.05	04:21 ± 00:34	04:20 ± 00:34	ns	01:09 ± 00:30	01:56 ± 0:58	< 0.01
Amplitude (°C)	< 0.001	ns	< 0.05	0.54 ± 0.11	0.61 ± 0.02	< 0.05	0.45 ± 0.09	0.44 ± 0.09	ns
Zenith (°C)	ns	< 0.01	< 0.01	37.19 ± 0.17	37.39 ± 0.12	< 0.001	37.21 ± 0.19	37.24 ± 0.19	ns
Nadir (°C)	< 0.001	< 0.05	ns	36.05 ± 0.17	36.10 ± 0.04	ns	36.29 ± 0.09	36.36 ± 0.28	ns
Mesor (°C)	< 0.05	< 0.05	< 0.01	36.65 ± 0.11	36.76 ± 0.10	< 0.001	36.75 ± 0.11	36.75 ± 0.14	ns

Values are expressed in means ± SD. Data are presented in local time. ns: non-significant.

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P300 amplitude may offer a good index of objective sleepiness and is stable in each individual.

Most of the studies that examined the effects of napping on alertness [1], cognitive performance [17], or motor imagery learning [13] were conducted in the elderly, shift workers or young healthy subjects, and all showed that napping, whatever the duration, was effective in enhancing cognitive performance. Those that evaluated alertness with P300 were less numerous and were conducted only in healthy subjects [32, 40, 47] or patients (for review see [18]) but never in athletes. Zhao et al. [47] and Milner et al. [32] showed in young habitual nappers that a nap, in normal sleep conditions, enhances vigilance, i. e. increases P300 amplitude and decreases P300 latency. According to these authors [47], the recuperative nature of short-term naps should be attributed to the delta activity during stage 2 sleep. Like these authors, we also found a significant amount of stage 2 (52.2 ± 20.82 %) during napping, which may have been sufficient to help restore the daytime vigilance level [21, 43] and lead to higher cortical arousal [47]. In addition, improvements in P300 components, found in our study, may be due to the nap itself, which took place on a favorable schedule, namely, during the post-lunch dip in vigilance, when core body temperature fell and sleep propensity increased [28]. The absence of sleep inertia after awakening may have been in favor of an increase of vigilance. Bastuji et al. [2] already reported that sleep inertia decreases P300 amplitude in both normal individuals and patients with excessive daytime sleepiness.

Assuming that the P300 amplitude reflects cognitive functioning in general, it might be suitable as an index for neurocognitive performance for athletes. Although the P300 remains fairly elusive with respect to the cognitive processes involved, there is broad consensus that the P300 indexes certain aspects of cognitive information processing [38]. Thus, the increase in P300 amplitude and the decrease in P300 latency mean that in our athletes, the amount of allocated attention resources is greater and the time required for the assessment of stimulation is decreased after the nap.

The improvements in the ERP components were accompanied by greater subjective alertness and psychomotor performance. These results demonstrate that a high level of vigilance is a necessary requirement to develop efficient attentional processes in athletic performance. Similar results were found by Waterhouse et al. [46] who reported positive effects on subjective alertness after a short post-lunch nap in partially sleep-deprived athletes. According to Campbell [9], there is a close relationship between P300 components and sleepiness. This relation was previously confirmed by Higushi et al. [23] who showed that P300 latency was positively correlated with subjective sleepiness and negatively correlated with subjective attention level.

In the shifted condition, only P300 amplitude on both Fz and Cz electrodes increased after the nap. We can assume that the restriction of sleep partly explained this result and that during the nap; the subjects had to restore this sleep deficit before using their skills to be efficient in neurophysiological processes. Furthermore, in contrast with the normal condition, the nap was carried out at an unfavorable schedule, when the body temperature rises. Higushi et al. [23] clearly reported a diurnal variation in P300 components, with longer latency and greater amplitude in the morning than in the afternoon. Moreover, according to Geisler and Polich [16], P300 components seem sensitive to physiological and psychological changes originating from individual differences in arousal level.

To date, no study has investigated the effects of simulated jet lag on P300 components. Nevertheless, a number of earlier studies demonstrated changes in P300 amplitude and/or P300 latency due to sleep deprivation or sleep irregularities. Takahashi and Arito [40] reported shorter P3 latency after a 15-min midday nap following a night of 4 h of sleep reduction in healthy subjects. Kingshott et al. [27] showed that P300 amplitude, recorded from the Fz and Cz electrodes, was reduced with sleep fragmentation, while P300 latency was not significantly increased. These discrepancies with our results may come from the composition of the previous sleep deprivation and of the presence or not of a subsequent nap.

In conclusion, this study shows the beneficial effects of a 20-min nap on P300 components, alertness and cognitive processing in athletes. These benefits are influenced by the circadian placement of the nap, duration of the nap and sleep inertia. This short nap, taken closer to the afternoon circadian dip in alertness, would be ideal for athletes whose cognitive performance is normally required immediately upon awakening. Thus, napping is a behavioral measure that may be used to counteract the deleterious effects of jet lag, alleviate a number of sleep-related problems, promote waking function and increase arousal.

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Conflict of Interest

The authors declare no conflict of interest.

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